MICROSCOPY

What is it?

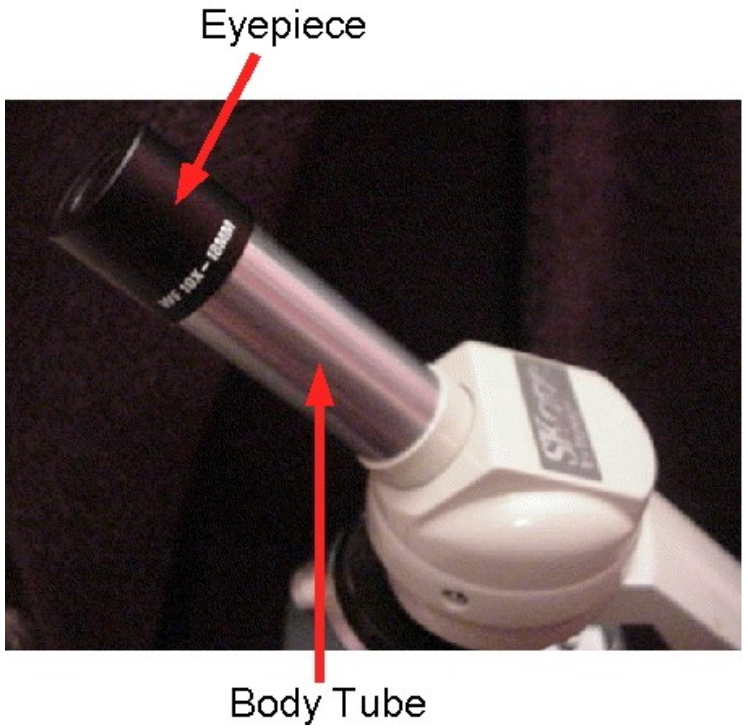
- The microscope pictured above is a compound light microscope.
- The term *light* refers to the method by which light transmits the image to your eye.
- C*ompound* deals with the microscope having more than one lens.
- *Microscope* is the combination of two words; "micro" meaning small and "scope" meaning view.



Parts and Uses

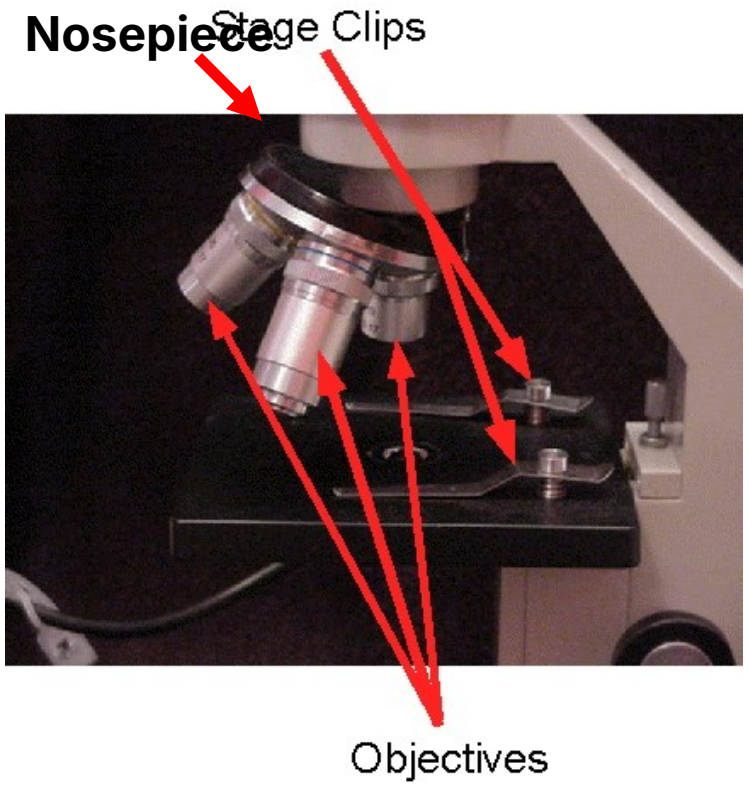
Eyepiece and Body Tube

- The *eyepiece* contains the ocular lens which magnifies objects a given amount that is listed on the eyepiece.
- The *body tube* supports the eyepiece and objectives.
- How much do our microscopes magnify?



Nosepiece, Objectives, and Stage Clips

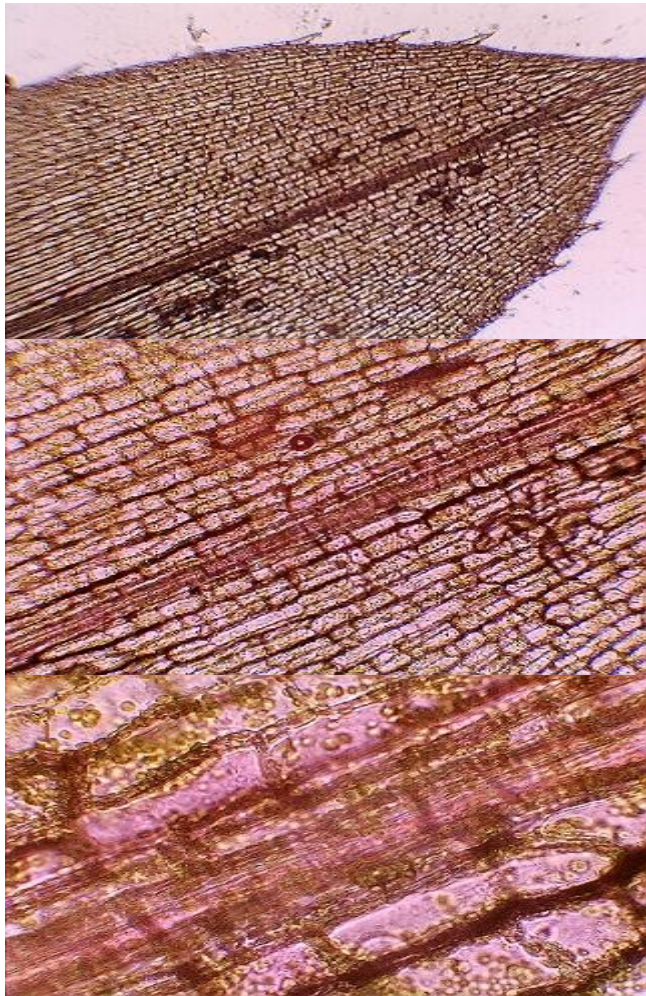
- The *nosepiece* holds the 3 objectives.
- The *objective lenses* range in magnification from 4X, 10X, and 40X.
- The *stage clips* holds the slide in place.



Magnification

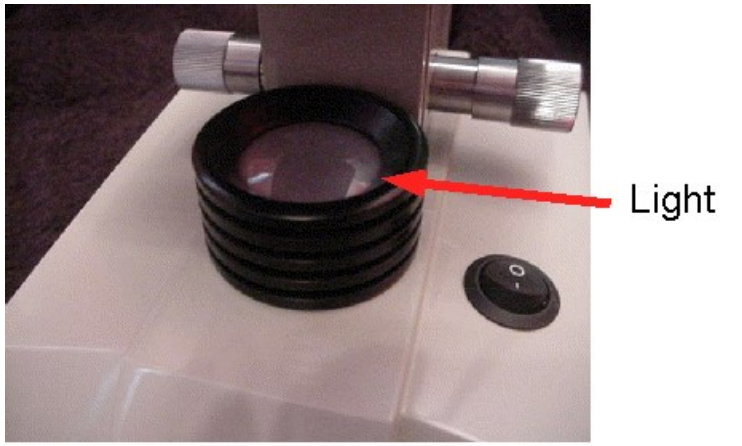
- What happens as the power of magnification increases?

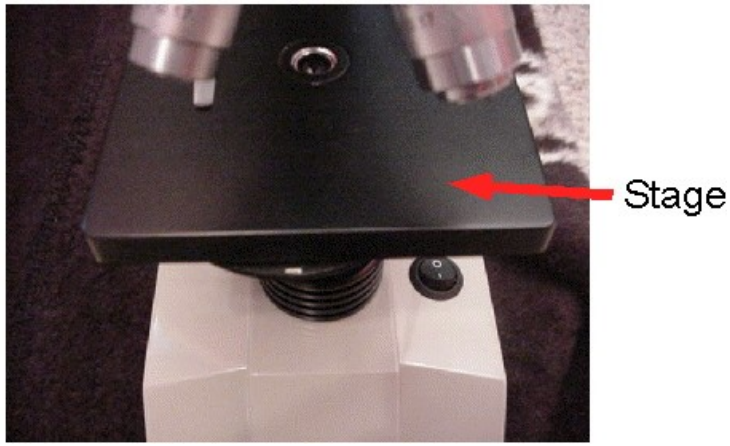




Stage, Light, and Diaphragm

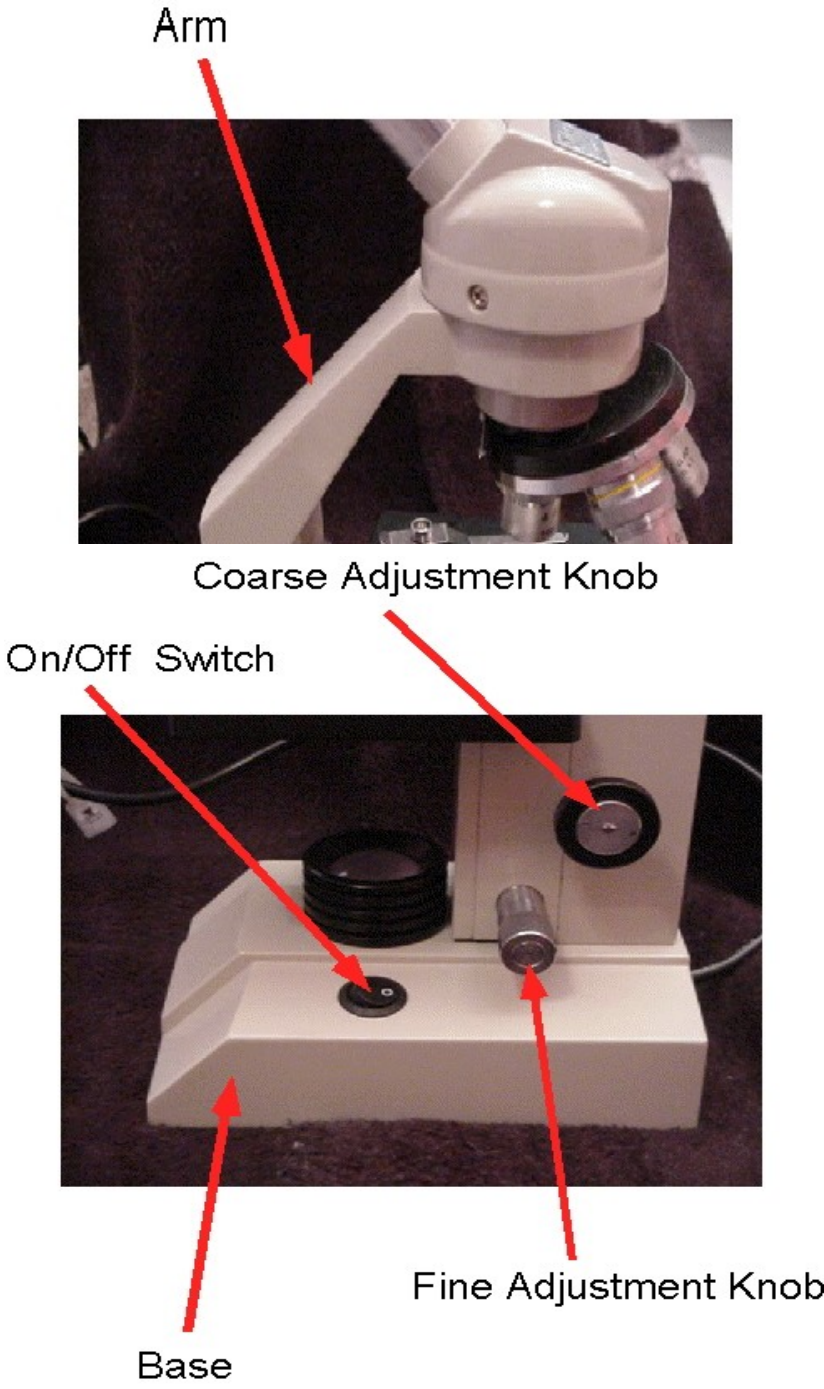
- The *stage* supports the slide being viewed.
- The *light source* projects upward through the diaphragm, the specimen, and the lenses.
- The *diaphragm* regulates the amount of light on the specimen.





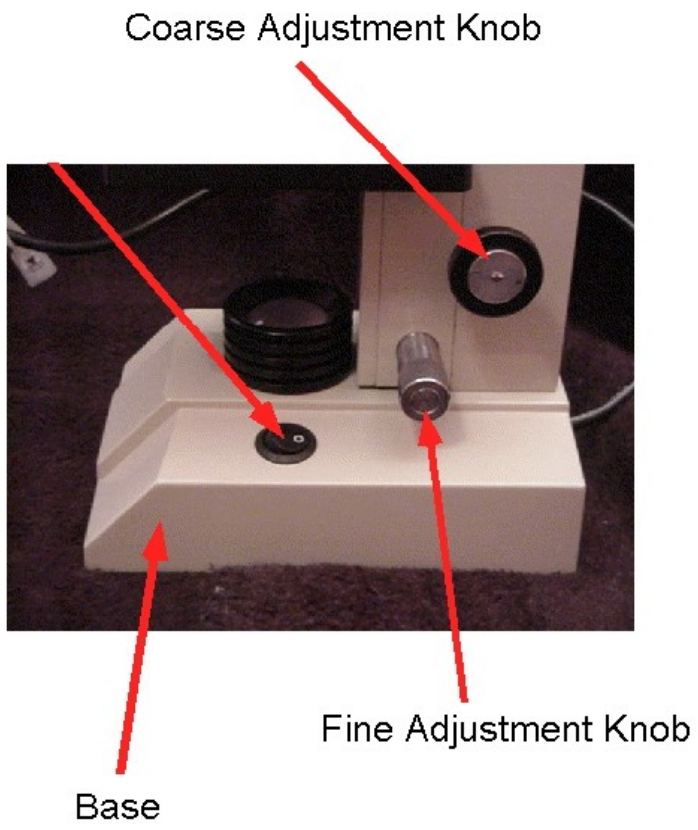
Arm and

- The *arm* is used to support the microscope when it is carried.
- The *base* supports the microscope.



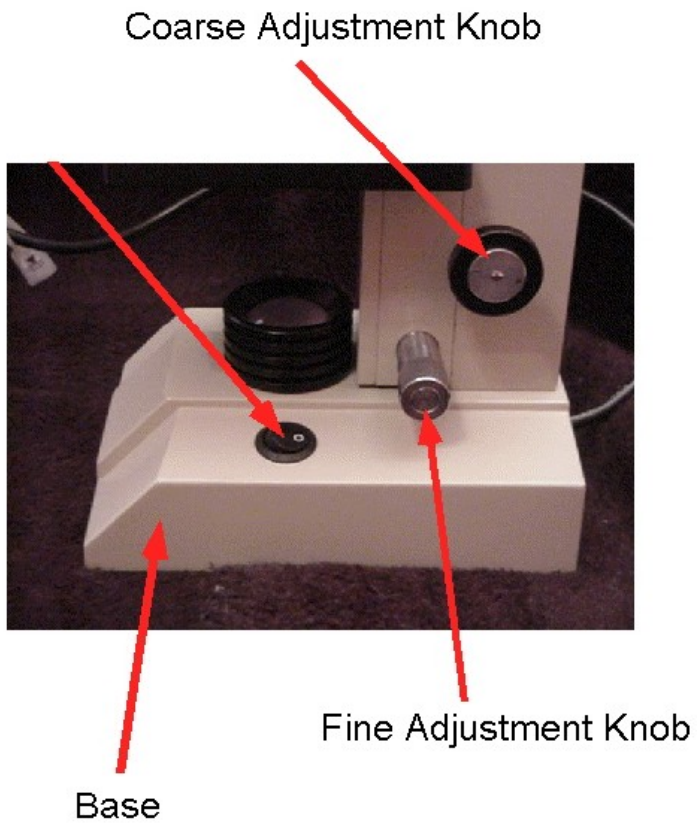
Coarse Adjustment Knob

- Moves the stage up and down for focusing.
- NOT TO BE USED WITH ANY OTHER OBJECTIVE BUT THE 4X



Fine Adjustment Knob

- Moves the stage slightly to sharpen the image.
- Used with the 10X and 40X objective to focus.



LIGHT / COMPOUND MICROSCOPE

- Based on the interaction of photons and tissue components.
- With light M/S stained preparations are usually examined by trans-illumination.
- Microscopic parts:
- A) Mechanical e.g stage, handle and nose piece etc
- B) Optical parts: 3 system of lenses.
- 1. condenser
- 2. objective
- 3. ocular

- The condenser collects and focus the light on object producing a cone of light.
- The objective produce enlarged image and project in the direction of ocular lens.
- The ocular lens further magnifies this image and project on viewers retina.

Magnification:

TM=Mag power of objective × Mag power of ocular

Resolution:

- A critical factor in obtaining crisp detailed image is resolution.
- The smallest distance between two particles at which the particles can be seen as separate objects.
- Quality of image, its clarity and richness of detail depends on the resolving power of M/C

Factors:

- Resolution depends on numerical aperture of objective lens.
- It also depends on wavelength of light.
- Magnification is independent of resolving part.
- Ocular lens enlarges the image but does not improve resolution.
- Resolving power of light M/C is 0.2 micrometer.

Types of microscope:

A) Using light source (photon)

B) Using electrons

Types of light M/C

1. Conventional
2. Phase contrast
3. Differential interference
4. Polarized
5. UV

Types of electron M/C

1. Scanning
2. Transmission

Phase contrast:

- Unstained biological specimen are usually transparent.
- All parts of specimen have same optical density.
- Principle :- light changes its speed and direction when passing through cellular and extracelluar structure with different RI.

These phase changes appear as light or dark relative to each other with this M/C

Differential interference:

- Produces three dimensional image of living cells and tissues (thickness can be measured)

Polarized M/C

- Polaroid filters are used which cause the light to vibrate in only one direction.
- Two filters used at Rt angle to each other. Summing effect-no light passes in eye piece.
- 1st filter below condenser, 2nd between objective and ocular lenses.
- Tissue containing oriented molecules such as cellulose, collagen, microtubules etc. Their repetitive molecular structure allow them to rotate the axis of light.
- These structures appear brighter against dark background and this ability is called Birefringence.

Electron Microscope:

Based on interaction of electrons and tissue component.

Principle: Beam of electron deflected by electromagnetic fields (similar to light).

- Electron produced by heating metallic filament (cathode) in a vacuum, forming a beam.
- Beam deflected by electromagnetic lenses. Image formed is seen on fluorescent screen or is projected on photographic plates.
- Heavy metals used to stain the section e.g uranium, lead.
- Resolution permitted 0.1 nm

- Working distance
- Field of vision
- Power of objective lenses